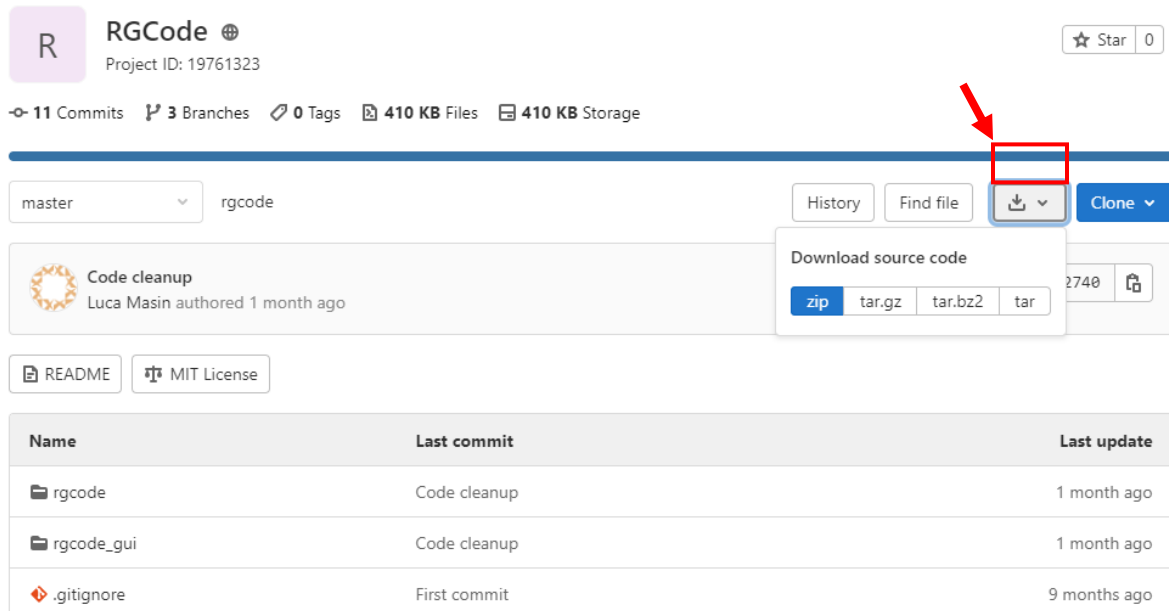
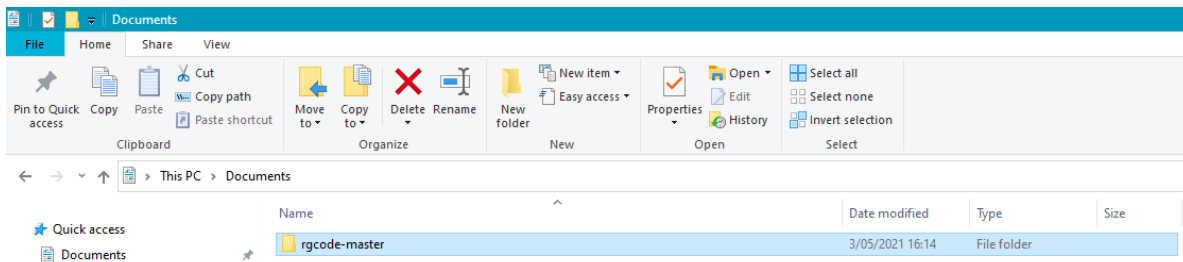


RGCode manual

1. Download and install the package manager [MiniConda](#). This allows RGCode to run inside a dedicated environment without interfering or causing conflicts with the host
2. Download the [RGCode from Gitlab](#). You will get an *rgcode-master.zip* file which you need to unzip

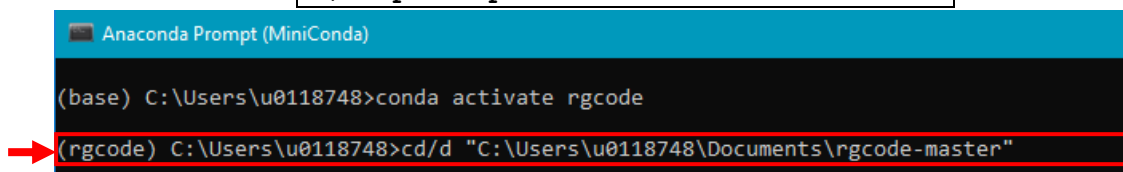


3. Open the Conda prompt, by looking for “Anaconda Prompt (Miniconda3)” in the start menu.
4. Go to the folder in which you unzipped *rgcode-master.zip* and find the *rgcode-master* folder. If you see the *setup.py* file inside, it is the right one. Select the folder and press “Copy path”, as seen below.



5. Move the prompt to the directory by running (type and press enter) the following command in the MiniConda prompt:

- For Windows users: `cd/d *paste path of RGCode-master folder*`



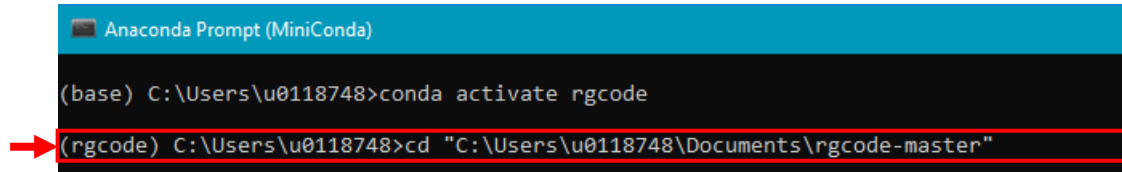
- For Mac users: `cd *paste path of RGCode-master folder*`

6. **Create an environment** by running (type and press enter) the following command in MiniConda:

```
conda env create -f rgcode.yml
```

7. **Activate the environment** by running (type and press enter) the following command in MiniConda:

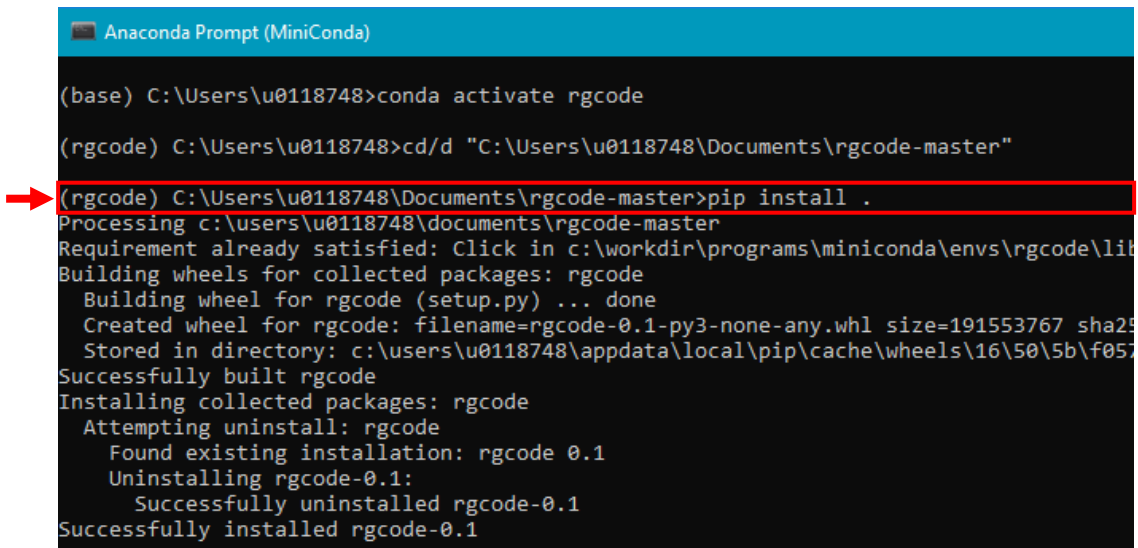
```
conda activate rgcode
```



```
Anaconda Prompt (MiniConda)
(base) C:\Users\u0118748>conda activate rgcode
(rgcode) C:\Users\u0118748>cd "C:\Users\u0118748\Documents\rgcode-master"
```

8. **Install packages** by running (type and press enter) the following command in MiniConda:

```
pip install .
```

 (Notice the dot at the end of the command)

```
Anaconda Prompt (MiniConda)
(base) C:\Users\u0118748>conda activate rgcode
(rgcode) C:\Users\u0118748>cd/d "C:\Users\u0118748\Documents\rgcode-master"
(rgcode) C:\Users\u0118748\Documents\rgcode-master>pip install .
Processing c:\users\u0118748\documents\rgcode-master
Requirement already satisfied: Click in c:\workdir\programs\miniconda\envs\rgcode\lib
Building wheels for collected packages: rgcode
  Building wheel for rgcode (setup.py) ... done
  Created wheel for rgcode: filename=rgcode-0.1-py3-none-any.whl size=191553767 sha256=
  Stored in directory: c:\users\u0118748\appdata\local\pip\cache\wheels\16\50\5b\f057
Successfully built rgcode
Installing collected packages: rgcode
  Attempting uninstall: rgcode
    Found existing installation: rgcode 0.1
    Uninstalling rgcode-0.1:
      Successfully uninstalled rgcode-0.1
Successfully installed rgcode-0.1
```

9. **Open the graphical interface** by running (type and press enter) the following command in MiniConda:

```
rgcode-gui
```

```
Anaconda Prompt (MiniConda) - rgcode-gui

(base) C:\Users\u0118748>conda activate rgcode

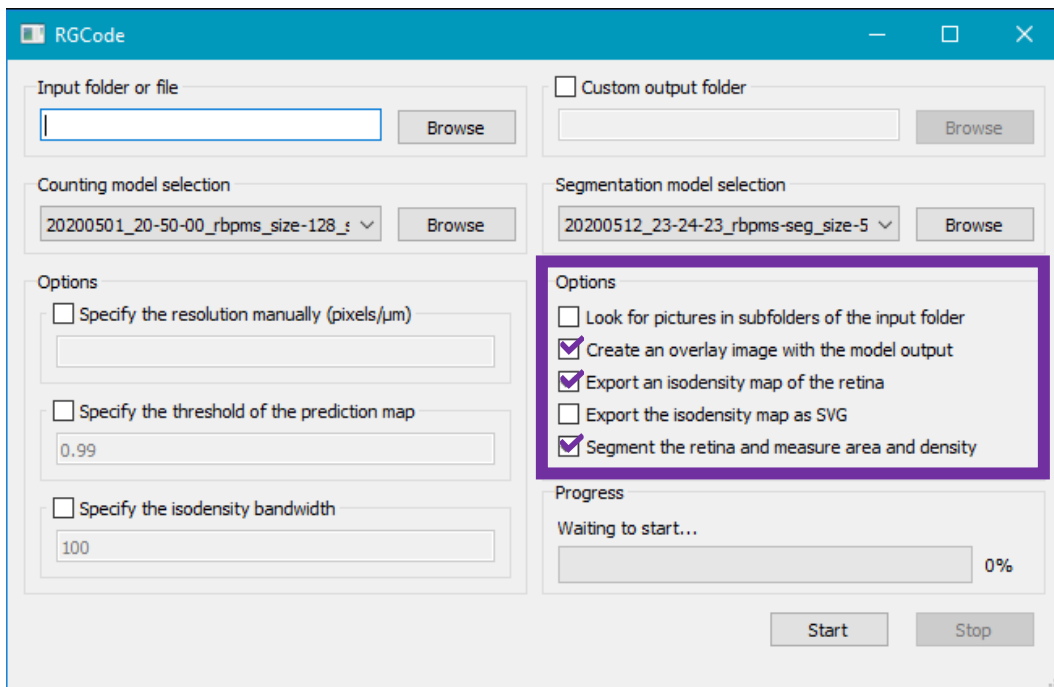
(rgcode) C:\Users\u0118748>cd/d "C:\Users\u0118748\Documents\rgcode-master"

(rgcode) C:\Users\u0118748\Documents\rgcode-master>pip install .
Processing c:\users\u0118748\documents\rgcode-master
Requirement already satisfied: Click in c:\workdir\programs\miniconda\envs\rgcode\lib\
Building wheels for collected packages: rgcode
  Building wheel for rgcode (setup.py) ... done
  Created wheel for rgcode: filename=rgcode-0.1-py3-none-any.whl size=191553767 sha256
  Stored in directory: c:\users\u0118748\appdata\local\pip\cache\wheels\16\50\5b\f0574
Successfully built rgcode
Installing collected packages: rgcode
  Attempting uninstall: rgcode
    Found existing installation: rgcode 0.1
    Uninstalling rgcode-0.1:
      Successfully uninstalled rgcode-0.1
Successfully installed rgcode-0.1

(rgcode) C:\Users\u0118748\Documents\rgcode-master>rgcode-gui
```

10. The graphical interface opens automatically. If you want to use RGCode as-is, simply specify the folder with the images of your RBPMS stained retinas and take a look at the options on the right (purple).

- You can specify a specific output folder. If left empty, the output folder can be found within the input folder.
- You can alter the counting and segmentation model, e.g. when you trained new models.
- When noticing counting/segmentation problems, you can manually specify the resolution / threshold of the prediction map / isodensity bandwidth. Or train a new model with your own data (more info [here](#))!



11. Press start, relax and wait till RGCode has fixed the job!

Made a mistake? Stop the script by pressing control + c.

12. Next time, repeat steps 7 and 9, skipping step 8.